Tutorial 7 Circular Motion Suggested Solutions

D1 **(a)** A resultant force acting on the particle is always perpendicular to the direction of its linear speed / velocity and is directed towards a fixed point. B1

The resultant force must be of constant magnitude. B1

(b) From the dynamical point of view, since the resultant (or centripetal) acceleration is perpendicular to the instantaneous velocity, the magnitude of the velocity will be unchanged hence the linear speed of such particle remains constant B1

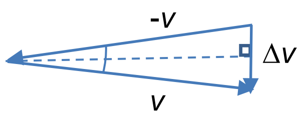
From the work-energy point of view, as the resultant force acts perpendicularly to the direction of motion, there is no displacement in the direction of the resultant force, hence work done by the resultant force is zero. This does not cause a change to the kinetic energy of the particle and hence its speed remains constant. B1

(c) Both the velocity and acceleration are not constant (changing continuously) as the direction of motion and the direction of the acceleration is constantly changing. B1

(Note that for uniform circular motion, while the magnitudes of the velocity and acceleration are constant, their directions are not. They are both vector quantities, and hence direction must be considered.)

D2 **(a)** Using sine rule,

$$\sin\frac{\theta}{2} = \frac{\Delta v}{2} \div v$$



$$\Delta v = 2 \times v \sin\frac{\theta}{2}$$

For small angles,

$$\sin\frac{\theta}{2} \approx \frac{\theta}{2}$$

$$\therefore \Delta v = 2v\left(\frac{\theta}{2}\right)$$

$$= v\theta$$

$$= 0.010v$$ A1

(b) Towards the centre of the circle. B1

(c) $s = r\theta$ M1

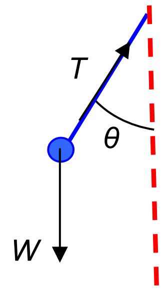
$$\Delta t = \frac{s}{v} = \frac{r\theta}{v} = \frac{0.010r}{v}$$ A1

(d) $$a = \frac{\Delta v}{\Delta t} = \frac{0.010v}{\left(\frac{0.010r}{v}\right)} = \frac{v^2}{r}$$ M1

D3 **(a)**

$T\sin\theta = ma_c$ --- (1)

$T\cos\theta = mg$ --- (2)



M1

$\frac{(1)}{(2)}$: $\tan\theta = \frac{a_c}{g}$

$\Rightarrow a_c = 9.81\times\tan15^o = 2.629 = 2.63\text{ m s}^{-2}$ A1

(b)

From $a_c = \frac{v^2}{r} = 2.629\text{ m s}^{-2}$ M1

$r = \frac{(23.0)^2}{2.629} = 201.2 = 201\text{ m}$ A1

(c)

From $\tan\theta = \frac{a_c}{g} = \frac{v^2}{rg}$ M1

$v = \sqrt{rg\tan\theta} = \sqrt{201.2\times9.81\times\tan9.0^\circ} = 17.68 = 17.7\text{ m s}^{-1}$ A1

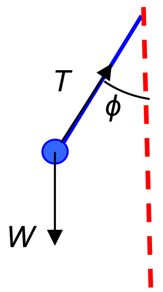
D4 **(a)**

$a_c = \frac{v^2}{r}$

$= \frac{(180)^2}{20000} = 1.62\text{ m s}^{-2}$ M1

A1

(b)



T – Tension A1

W – Weight of pendulum bob A1

(c) Horizontally to the right B1

(d)

$T\sin\phi = \frac{mv^2}{r}$ --(1)

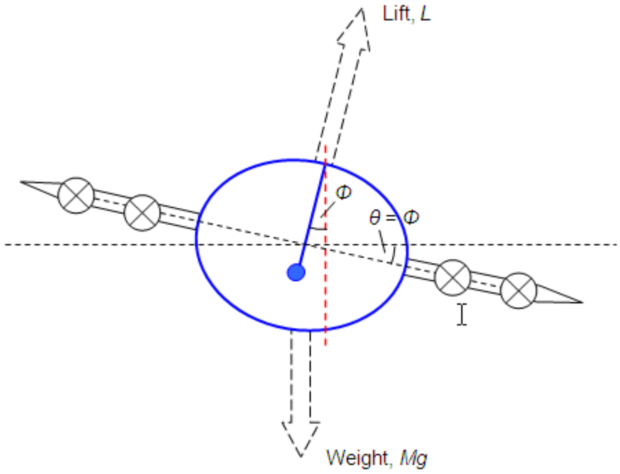
$T\cos\phi = mg$ --(2) M1

$(1)/(2)$,

$\tan\phi = \frac{v^2}{rg}$

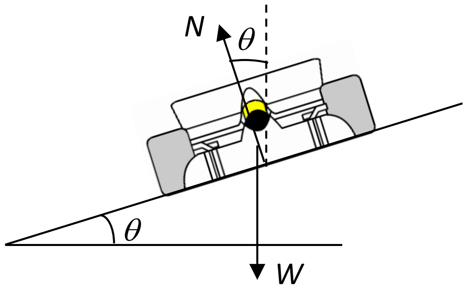
$\phi = \tan^{-1}\left(\frac{180^2}{20000\times9.81}\right) = 9.4^0$ A1

(e) B1



D5 **(a)** The centripetal force is provided purely by the horizontal component of the normal contact force N which is directed towards the centre of the curve

f = 0 N



Vertically: $N\cos\theta = mg$ ... (1)

Horizontally: $N\sin\theta = \dfrac{mv^2}{r}$ ... (2) M1

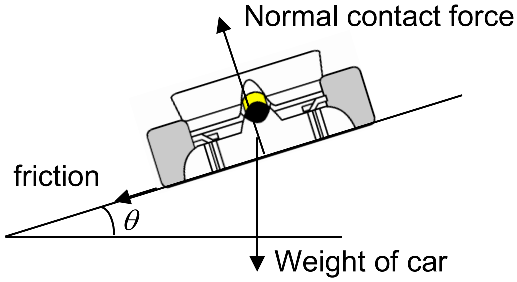
$\dfrac{(2)}{(1)}: \theta = \tan^{-1}\left(\dfrac{v^2}{rg}\right) = \tan^{-1}\left(\dfrac{\left(\dfrac{80000}{60\times 60}\right)^2}{(200)(9.81)}\right) = 14.1°$ A1

(b) The car will still be able to negotiate the bend. B1

Although the centripetal force required to turn the car increases with the mass of the car, the horizontal component of the normal force also increases proportionately with the increase in mass, hence the horizontal component of the normal force is still able to provide the centripetal force. B1

*For ideal banking, the centripetal force does not depend on the friction between the road and the tires.

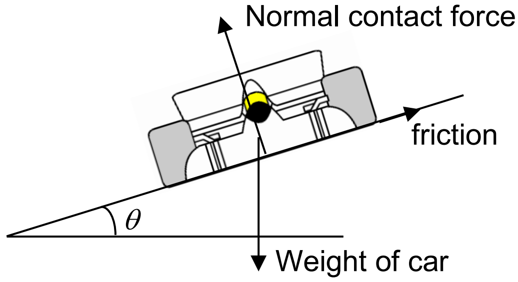
(c) **(i)**



Vertically: $N\cos\theta = mg + f\sin\theta$

Horizontally: $N\sin\theta + f\cos\theta = \dfrac{mv^2}{r}$ B2

(ii) B2



Vertically: $N\cos\theta + f\sin\theta = mg$

Horizontally: $N\sin\theta - f\cos\theta = \dfrac{mv^2}{r}$

D6 (a) The pilot feels the lightest at the top of his flight as the centripetal force on him is provided by both the normal contact force of the seat on him and his weight. B1

The pilot feels the heaviest at the bottom of his flight as the centripetal force on him is the difference between the normal contact force and his weight. For a uniform circular motion, the centripetal force term is the same and weight remaining constant, the normal contact force must, at this position be higher than at any other positions in the circular path. Hence, the pilot feels that he is heaviest there. B1

(b) At the top of the circle, $N + W = \dfrac{mv^2}{r}$, where N = normal contact force by head on hat, W = hat's weight B1

When the linear speed at the top is high enough such that the centripetal force, $\dfrac{mv^2}{r}$ is greater than W, N will be positive. This implies the hat remains intact. B1

If the linear speed is too small such that the centripetal force, $\dfrac{mv^2}{r}$ is lower than W, N will be negative. Physically, the normal contact force can only be zero or positive. Hence N will be zero, which means that the hat will fall off. B1

(c) (i)

$$a_c = \frac{v^2}{r}$$

$$= \frac{(12)^2}{7.0} = 21 \text{ m s}^{-2}$$ M1 A1

(ii)

$$W + N = \frac{mv^2}{r}$$ M1

$$\therefore \text{ N} = \frac{mv^2}{r} - mg = 60(20.571 - 9.81) = 646 \text{ N}$$ A1

(d) (i)

$$\Delta U = -mgh$$ M1

$$= -60 \times 9.81 \times 14$$

$$= -8240 \text{ J}$$ A1

(ii) Loss in GPE = Gain in KE of the passenger

$$mgh = \frac{1}{2}mv_B^2 - \frac{1}{2}mv_T^2$$ M1

$$8240.4 = \frac{1}{2}(60)v_B{}^2 - \frac{1}{2}(60)(12)^2$$

$$v_B = 20.5 \text{ m s}^{-1}$$ A1

(e) This is so that the normal contact force on the passengers is positive and they remain in their seats at the top of the loop. B1

At the top,

$$R + mg = \frac{mv^2}{r}$$

$$R \geq 0 \quad \Rightarrow \quad \frac{mv^2}{r} - mg \geq 0 \quad \Rightarrow \quad v \geq \sqrt{rg}$$ M1

minimum v at the top, $v_{top} = \sqrt{rg} = \sqrt{7.0 \times 9.81} =$ 8.287 m s^{-1} M1

From bottom to top, using principle of conservation of energy,
increase in G.P.E. = decrease in K.E.

$$mgh = \frac{1}{2}mv^2{}_{bottom} - \frac{1}{2}mv^2{}_{top}$$

$$\therefore v_{bottom} = \sqrt{2gh + v^2{}_{top}} = \sqrt{2 \times 9.81 \times 14 + (8.287)^2}$$ M1

$$= 18.5 \text{ m s}^{-1}$$ A1

D7 **(a)** At C,

$$T + mg = \frac{mv^2}{L}$$ M1

If string is just to be taut T = 0, M1

$$\therefore \therefore v = \sqrt{gL}$$

(b) decrease in KE = increase in GPE of the particle M1

$$\frac{1}{2}mv_B^2 - \frac{1}{2}mv_C^2 = mg(2L)$$

$$v_B^2 - v_C^2 = 4gL$$

$$v_B^2 = 4gL + gL = 5gL$$ A1

$$\therefore v_B = \sqrt{5gL}$$

(c) The string is most likely to break when the particle is at the lowest position of the circular path as tension of string is greatest at the bottom. B1 B1

D8 **(a)** **(i)** The required centripetal force is given by the sum of the tension due to the rod and the weight of the ball, which are both acting in the same direction in this case. B1

Hence, the centripetal force is more than 2mg.

(ii)

$$F_C = T + mg$$

$$= 2mg + mg = 3mg$$ B1

(iii) When the ball is directly below C, the tension in the rod and the weight of the ball are in opposite directions. Since the same centripetal force is required in uniform circular motion,

$$F_C = T_{bottom} - mg = 3mg$$ A1

$$T_{bottom} = 4mg$$

(b) **(i)** $F = mr\omega^2 = 3mg$ M1

$0.72\omega^2 = 3 \times 9.81$ C1

$\omega = 6.393 = 6.39 \text{ rad s}^{-1}$ A1

(ii) $v = r\omega$

$= (0.72)(6.393)$ M1

$= 4.603 = 4.60 \text{ m s}^{-1}$ A1

(c) **(i)** Although the ball loses gravitational potential energy due to a loss in height, its kinetic energy remains constant. B1

Therefore, the friction between moving parts/gears at the pivot C does work against the ball-rod system to maintain constant kinetic energy or speed. B1

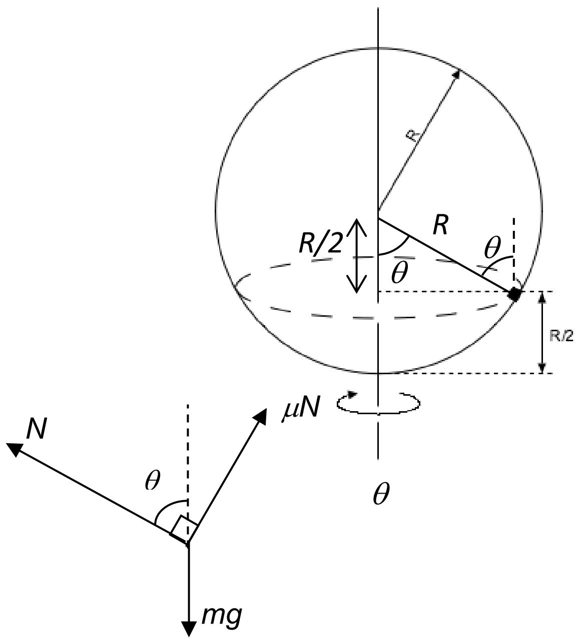
(ii) Work done $=$ Change in G.P.E

$= mg(\Delta h)$

$= (0.240 \times 9.81) \times (-0.72 \times 2)$ M1

$= -3.39 \text{ J}$ A1

C1 **(a)** From diagram, $\cos\theta = \dfrac{R/2}{R} \quad \Rightarrow \theta = 60°$



Resolving forces vertically,
$\mu N\cos 30° + N\sin 30° = mg$

$$\Rightarrow N\left(\frac{\sqrt{3}}{2}\mu + \frac{1}{2}\right) = mg \quad \text{———— (1)}$$

Resolving forces horizontally,
$N\cos 30° - \mu N\cos 60° = mr\omega^2$

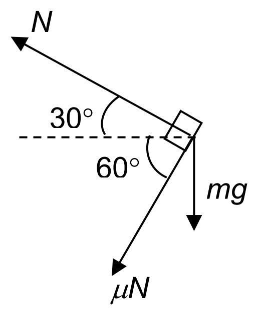
$= mR\sin 60°\,\omega^2$

$$\Rightarrow N\left(\frac{\sqrt{3}}{2} - \frac{\mu}{2}\right) = \frac{\sqrt{3}}{2}mR\omega^2 \quad \text{———— (2)}$$

Solving (1) and (2), we obtain $\underline{\mu = 0.216}$

(b) When $\omega = 8 \text{ rad s}^{-1}$, the friction acts downward as object tends to move upwards.

Hence, the vector diagram for the forces is as follows:



Once again, we obtain two equations:

$$N\left(1 - \sqrt{3}\mu\right) = 2mg \quad \text{——————— (3)}$$

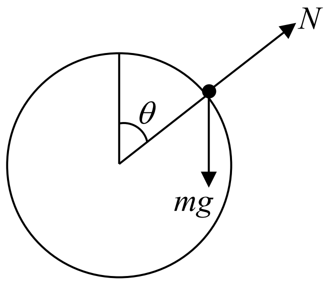
$$N\left(\frac{\sqrt{3}}{2} + \frac{1}{2}\mu\right) = \frac{\sqrt{3}}{2}mR\omega^2 \quad \text{——————- (4)}$$

Solving, we get $\underline{\mu = 0.185}$

C2 Resolving forces in the radial direction,

$mg\cos\theta - N = mv^2/r$

When the bead breaks contact, $N = 0$.



Hence, $mg\cos\theta = mv^2/r$

$\cos\theta = v^2/gr$

By the principle of conservation of energy,

$$\frac{1}{2}mv^2 = mgr(1-\cos\theta)$$

$$v^2/gr = 2(1-\cos\theta)$$

Hence, $\cos\theta = 2/3 \Rightarrow \theta = \cos^{-1}(2/3)$.